



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.NOS.4

Prime Factorization

Lesson 6-13 • Teacher Copy

Name: _____

Date: _____

Class: _____

ORBITAL LOGISTICS MISSION

Cargo Codebreak

You are the logistics officer aboard Station Helios. A shipment of 60 supply crates just docked, and the sorting robots can only distribute cargo once it is broken into its prime building blocks. Master prime factorization and the station eats this week.



TODAY'S OBJECTIVES

Content Objective: I can write a number as a product of its prime factors using a factor tree.

Language Objective: I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Prime Factorization

Prime number

Composite number

Factor

Factor tree

Exponent



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Writing a number as numbers multiplied together is its prime factorization.

2

A whole number greater than 1 with exactly two factors, 1 and itself, is a number.

3

A whole number greater than 1 that has more than two factors is a number.

4

A number that divides evenly into another number is called a .

5 I can break a number into its prime factors step by step using a

6 In 2^3 , the small number 3 is the and it shows 2 is multiplied 3 times.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Starting from its prime factorization, how can the gardener use the factors of 72 to decide which rectangular grids will use all 72 seed pods?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Prime Factorization** — Writing a whole number as a product of only prime numbers.

Factorización prima — Escribir un número entero como un producto de solo números primos.

- ☐ **Prime number** — A number bigger than 1 that you can only divide by 1 and itself.

Número primo — Un número mayor que 1 que solo se puede dividir entre 1 y sí mismo.

- ☐ **Composite number** — A number bigger than 1 that you can divide by more than just 1 and itself.

Número compuesto — Un número mayor que 1 que se puede dividir entre más números, no solo 1 y sí mismo.

- ☐ **Factor** — A factor of a whole number divides it evenly, with no remainder. For example, 3 is a factor of 12 because $12 \div 3 = 4$.

Factor — Un factor de un número entero lo divide exactamente, sin dejar residuo. Por ejemplo, 3 es un factor de 12 porque $12 \div 3 = 4$.

- ☐ **Factor tree** — A picture that splits a number into its prime numbers, step by step.

Árbol de factores — Un dibujo que separa un número en sus números primos, paso a paso.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

Because $72 = \underline{\hspace{1cm}}$, I can multiply combinations of those prime factors to get factors like $\underline{\hspace{1cm}}$, so a grid that works is $\underline{\hspace{1cm}}$ by $\underline{\hspace{1cm}}$.

Di tu respuesta primero: Mi respuesta es $\underline{\hspace{1cm}}$ porque $\underline{\hspace{1cm}}$.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: 60 is composite because it has more than two factors.

Evidence: Students say 60 is composite because it has more than two factors (it is divisible by numbers other than 1 and 60, such as 2, 3, 5, 6), so it can be broken down.

Reasoning: This evidence connects the definition of prime factorization to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Because** $72 = \underline{\hspace{1cm}}$, **I can multiply combinations of those prime factors to get factors like** $\underline{\hspace{1cm}}$, **so a grid that works is** $\underline{\hspace{1cm}}$ **by** $\underline{\hspace{1cm}}$.
- **My answer is** $\underline{\hspace{1cm}}$.

Mi respuesta es $\underline{\hspace{1cm}}$.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is** $\underline{\hspace{1cm}}$.
- **I know because** $\underline{\hspace{1cm}}$.

Mi afirmación es $\underline{\hspace{1cm}}$.

Lo sé porque $\underline{\hspace{1cm}}$.

- **So** $\underline{\hspace{1cm}}$.

Entonces $\underline{\hspace{1cm}}$.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Prime Factorization
2. **Notes 2:** Prime number
3. **Notes 3:** Composite number
4. **Notes 4:** Factor
5. **Notes 5:** Factor tree
6. **Notes 6:** Exponent
7. **Watch for this mistake:** *A common mistake in Prime Factorization is stopping the factor tree too soon — leaving a composite number like 4, 6, 8, or 9 in the final answer instead of continuing to split it. For example, writing $20 = 4 \times 5$ as the prime factorization (4 is composite) instead of finishing with $20 = 2 \times 2 \times 5$. Before you submit, check every number in your final answer: if any factor can still be broken into smaller factors, the tree isn't finished yet.*
8. **Try It 1:** A. $36 - 2^2 \times 3^2 = 4 \times 9 = 36$, so 36 is the code for this panel.
9. **Try It 2:** A. $2 \times 3 \times 13 - 78 = 2 \times 39 = 2 \times 3 \times 13$. All three factors are prime, so $2 \times 3 \times 13$ unlocks the final panel.